

10th grade

Learning evidence 2.1: Analysis of Decisions (Practice Solutions)

Evaluated criteria

This exam evaluates the following criteria. Each criterion is evaluated across the items below. A student passes a criterion if it is correctly applied in the solution.

- C2 Interprets decision alternatives, events, consequences, and states of nature.
- C3 Builds a payoff table from a description of the problem.
- C4 Explains the maximax criterion for decision making without probabilities.
- C5 Summarizes the maximin criterion for decision making without probabilities.
- C1 Describes how a problem can be formulated for optimal decision making, computes the expected value, and interprets the result.

Problems

P1 Problem description

A community recreation center must choose a membership model for the next year: Model A (standard monthly pass), Model B (premium pass with classes), or Model C (flex pass). Demand for memberships can be high, medium, or low, and staffing costs can be favorable or unfavorable. The demand probabilities are 0,30 (high), 0,45 (medium), and 0,25 (low). Staffing costs are independent of demand, with probabilities 0,55 (favorable) and 0,45 (unfavorable). Expected membership counts depend on demand and model: under high demand the center expects 640 members with Model A, 500 with Model B, and 720 with Model C; under medium demand it expects 480 with Model A, 380 with Model B, and 540 with Model C; under low demand it expects 320 with Model A, 250 with Model B, and 380 with Model C. Membership fees are 36 for Model A, 50 for Model B, and 32 for Model C per member per year. Variable staffing cost per member is 19 when costs are favorable and 27 when costs are unfavorable. Fixed annual costs are 6,800 for Model A, 9,200 for Model B, and 5,400 for Model C. All monetary amounts in this problem can be treated as dollars. Use the combined demand and staffing outcomes as states of nature and the stated probabilities to compute expected profits.

C2

Decision Alternatives	Model A (standard), Model B (premium), Model C (flex)
States of Nature	Demand high, medium, low with staffing costs favorable or unfavorable
Events	Realized demand level paired with staffing cost conditions in the year
Consequences	Annual profit in dollars from membership fees minus staffing and fixed costs

C3

Payoff = revenue – cost, where revenue = (members × fee) and cost = (members × variable cost) + fixed cost. *Members per model by state of nature*

Model	High demand	Medium demand	Low demand
Model A	640	480	320
Model B	500	380	250
Model C	720	540	380

Cost parameters by model

Model	Favorable cost	Unfavorable cost
Model A	19	27
Model B	19	27
Model C	19	27

Revenue parameters by model

Model	High	Medium	Low
Model A	36	36	36
Model B	50	50	50
Model C	32	32	32

State of nature	Model A revenue	Model B revenue	Model C revenue
High demand, favorable cost	$640 \cdot 36$ $= 23040$	$500 \cdot 50$ $= 25000$	$720 \cdot 32$ $= 23040$
High demand, unfavorable cost	$640 \cdot 36$ $= 23040$	$500 \cdot 50$ $= 25000$	$720 \cdot 32$ $= 23040$
Medium demand, favorable cost	$480 \cdot 36$ $= 17280$	$380 \cdot 50$ $= 19000$	$540 \cdot 32$ $= 17280$
Medium demand, unfavorable cost	$480 \cdot 36$ $= 17280$	$380 \cdot 50$ $= 19000$	$540 \cdot 32$ $= 17280$
Low demand, favorable cost	$320 \cdot 36$ $= 11520$	$250 \cdot 50$ $= 12500$	$380 \cdot 32$ $= 12160$
Low demand, unfavorable cost	$320 \cdot 36$ $= 11520$	$250 \cdot 50$ $= 12500$	$380 \cdot 32$ $= 12160$

State of nature	Model A cost	Model B cost	Model C cost
High demand, favorable cost	$640 \cdot 19 + 6800$ $= 18960$	$500 \cdot 19 + 9200$ $= 18700$	$720 \cdot 19 + 5400$ $= 19080$
High demand, unfavorable cost	$640 \cdot 27 + 6800$ $= 24080$	$500 \cdot 27 + 9200$ $= 22700$	$720 \cdot 27 + 5400$ $= 24840$
Medium demand, favorable cost	$480 \cdot 19 + 6800$ $= 15920$	$380 \cdot 19 + 9200$ $= 16420$	$540 \cdot 19 + 5400$ $= 15660$
Medium demand, unfavorable cost	$480 \cdot 27 + 6800$ $= 19760$	$380 \cdot 27 + 9200$ $= 19460$	$540 \cdot 27 + 5400$ $= 19980$
Low demand, favorable cost	$320 \cdot 19 + 6800$ $= 12880$	$250 \cdot 19 + 9200$ $= 13950$	$380 \cdot 19 + 5400$ $= 12620$
Low demand, unfavorable cost	$320 \cdot 27 + 6800$ $= 15440$	$250 \cdot 27 + 9200$ $= 15950$	$380 \cdot 27 + 5400$ $= 15660$

Profit is computed as Revenue minus Cost for each alternative and state.

State of nature	Model A	Model B	Model C
High demand, favorable cost	23040 – 18960 = 4080	25000 – 18700 = 6300	23040 – 19080 = 3960
High demand, unfavorable cost	23040 – 24080 = –1040	25000 – 22700 = 2300	23040 – 24840 = –1800
Medium demand, favorable cost	17280 – 15920 = 1360	19000 – 16420 = 2580	17280 – 15660 = 1620
Medium demand, unfavorable cost	17280 – 19760 = –2480	19000 – 19460 = –460	17280 – 19980 = –2700
Low demand, favorable cost	11520 – 12880 = –1360	12500 – 13950 = –1450	12160 – 12620 = –460
Low demand, unfavorable cost	11520 – 15440 = –3920	12500 – 15950 = –3450	12160 – 15660 = –3500

Combined probabilities are $0,30 \cdot 0,55 = 0,165$, $0,30 \cdot 0,45 = 0,135$, $0,45 \cdot 0,55 = 0,2475$, $0,45 \cdot 0,45 = 0,2025$, $0,25 \cdot 0,55 = 0,1375$, $0,25 \cdot 0,45 = 0,1125$.

State of nature	Probability	Model A	Model B	Model C
High demand, favorable cost	0.165	4080	6300	3960
High demand, unfavorable cost	0.135	-1040	2300	-1800
Medium demand, favorable cost	0.2475	1360	2580	1620
Medium demand, unfavorable cost	0.2025	-2480	-460	-2700
Low demand, favorable cost	0.1375	-1360	-1450	-460
Low demand, unfavorable cost	0.1125	-3920	-3450	-3500

C4

$$\text{máx}(\text{Model A}) = \text{máx}\{4080, -1040, 1360, -2480, -1360, -3920\} = 4080,$$

$$\text{máx}(\text{Model B}) = \text{máx}\{6300, 2300, 2580, -460, -1450, -3450\} = 6300,$$

$$\text{máx}(\text{Model C}) = \text{máx}\{3960, -1800, 1620, -2700, -460, -3500\} = 3960.$$

$$\text{máx}\{\text{máx}(\text{Model A}), \text{máx}(\text{Model B}), \text{máx}(\text{Model C})\} = \text{máx}\{4080, 6300, 3960\} = 6300.$$

The Maximax choice is Model B with 6300. The criterion is *risky* because it focuses only on the best possible payoff and ignores how likely it is.

C5

$$\text{mín}(\text{Model A}) = \text{mín}\{4080, -1040, 1360, -2480, -1360, -3920\} = -3920,$$

$$\text{mín}(\text{Model B}) = \text{mín}\{6300, 2300, 2580, -460, -1450, -3450\} = -3450,$$

$$\text{mín}(\text{Model C}) = \text{mín}\{3960, -1800, 1620, -2700, -460, -3500\} = -3500.$$

$$\text{máx}\{\text{mín}(\text{Model A}), \text{mín}(\text{Model B}), \text{mín}(\text{Model C})\} = \text{máx}\{-3920, -3450, -3500\} = -3450.$$

The Maximin choice is Model B with -3450 because it has the least severe worst-case loss. This is *safe* because it protects against the worst outcome.

C1

Expected values use the combined-state probabilities.

$$\begin{aligned}
 EV_A &= 0,165(4080) + 0,135(-1040) + 0,2475(1360) + 0,2025(-2480) + 0,1375(-1360) + 0,1125(-3920) \\
 &= 673,2 - 140,4 + 336,6 - 502,2 - 187 - 441 = -260,8, \\
 EV_B &= 0,165(6300) + 0,135(2300) + 0,2475(2580) + 0,2025(-460) + 0,1375(-1450) + 0,1125(-3450) \\
 &= 1039,5 + 310,5 + 638,55 - 93,15 - 199,375 - 388,125 = 1307,9, \\
 EV_C &= 0,165(3960) + 0,135(-1800) + 0,2475(1620) + 0,2025(-2700) + 0,1375(-460) + 0,1125(-3500) \\
 &= 653,4 - 243 + 400,95 - 546,75 - 63,25 - 393,75 = -192,4.
 \end{aligned}$$

The expected value criterion chooses Model B because 1307,9 is the largest expected profit. This aligns with both the Maximax and Maximin outcomes for Model B, so it is the best overall decision under uncertainty.

P2 Problem description

A school club must choose one of three snack plans for a two-week fundraiser: Plan A (fresh-made items), Plan B (mixed menu), or Plan C (budget staples). Demand for snacks can be high, medium, or low. Based on past fundraisers, the probability of high demand is 0,30, medium demand is 0,50, and low demand is 0,20. Profits are measured in hundreds of dollars. If demand is high, Plan A yields 88, Plan B yields 74, and Plan C yields 60. If demand is medium, Plan A yields 54, Plan B yields 66, and Plan C yields 50. If demand is low, Plan A yields -6 , Plan B yields 28, and Plan C yields 36. The stated probabilities must be used for an expected value comparison as part of the decision analysis.

C2

Decision Alternatives	Plan A (fresh-made items), Plan B (mixed menu), Plan C (budget staples)
States of Nature	Demand high, medium, low with probabilities 0,30, 0,50, 0,20
Events	Realized demand level in the serving period
Consequences	Profit in hundreds of dollars after all costs

C3

Payoff = revenue – cost, and the problem already provides the resulting profits for each plan and demand level.

$$\begin{aligned}
 &\text{High demand: } \Pi_A = 88, \Pi_B = 74, \Pi_C = 60, \\
 &\text{Medium demand: } \Pi_A = 54, \Pi_B = 66, \Pi_C = 50, \\
 &\text{Low demand: } \Pi_A = -6, \Pi_B = 28, \Pi_C = 36.
 \end{aligned}$$

State of nature	Probability	Plan A	Plan B	Plan C
High demand	0.30	88	74	60
Medium demand	0.50	54	66	50
Low demand	0.20	-6	28	36

C4

$$\text{máx}(\text{Plan A}) = \text{máx}\{88, 54, -6\} = 88,$$

$$\text{máx}(\text{Plan B}) = \text{máx}\{74, 66, 28\} = 74,$$

$$\text{máx}(\text{Plan C}) = \text{máx}\{60, 50, 36\} = 60.$$

$$\text{máx}\{\text{máx}(\text{Plan A}), \text{máx}(\text{Plan B}), \text{máx}(\text{Plan C})\} = \text{máx}\{88, 74, 60\} = 88.$$

The Maximax choice is Plan A. It is *risky* because it only considers the highest possible profit.

C5

$$\text{mín}(\text{Plan A}) = \text{mín}\{88, 54, -6\} = -6,$$

$$\text{mín}(\text{Plan B}) = \text{mín}\{74, 66, 28\} = 28,$$

$$\text{mín}(\text{Plan C}) = \text{mín}\{60, 50, 36\} = 36.$$

$$\text{máx}\{\text{mín}(\text{Plan A}), \text{mín}(\text{Plan B}), \text{mín}(\text{Plan C})\} = \text{máx}\{-6, 28, 36\} = 36.$$

The Maximin choice is Plan C with 36 because it gives the best worst-case outcome. It is *safe* because it prioritizes the least harmful result.

C1

$$EV_A = 0,30(88) + 0,50(54) + 0,20(-6) = 26,4 + 27 - 1,2 = 52,2,$$

$$EV_B = 0,30(74) + 0,50(66) + 0,20(28) = 22,2 + 33 + 5,6 = 60,8,$$

$$EV_C = 0,30(60) + 0,50(50) + 0,20(36) = 18 + 25 + 7,2 = 50,2.$$

The expected value criterion chooses Plan B because 60,8 is the largest expected profit. This differs from the Maximax choice (Plan A) and the Maximin choice (Plan C), so the expected value gives a balanced decision using the probabilities.